

CO2-AT1:

CSA0905-Java programming:

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ENGINEERING

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JAVA PROGRAMMING

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Smart Document Editor using String, StringBuffer and Scanner

Q1 Smart Document Editor using String, StringBuffer and Scanner
10 marks

Q2 Hospital Token Management System using List Interface
10 marks

Q3 Airport Boarding Queue Management using Queue Interface
10 marks

Q4 Digital Certificate Verification using Set Interface
10 marks

Q5 Smart Inventory Management using Map Interface
10 marks

5/5 answered

Question 1 :

- 1.Read a paragraph from the user.
- 2.Count the number of characters, words, and sentences.
- 3.Replace every occurrence of a given word with another word.
- 4.Reverse each sentence individually.
- 5.Maintain editing history using StringBuffer.
- 6.Construct the final formatted document using StringBuffer.
- 7.Display the original document, editing history, and final document.

Requirement: Use String methods wherever appropriate. Use StringBuffer for document construction and StringBuffer for edit history.

Your Code

```
1 import java.util.*;
2
3 public class SmartDocumentEditor{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6
7         String paragraph=sc.nextLine();
8
9         int characters=paragraph.length();
10
11         String words[]=paragraph.trim().split("\\s+");
12         int wordCount=words.length;
13
14         String sentences[]=paragraph.split("[.!?]");
15         int sentenceCount=0;
16
17         for(String s:sentences){
18             if(!s.trim().isEmpty()){
19                 sentenceCount++;
```

Input

Java is easy. Java is powerful.
Java
Python

Output

```
Original Document:
Java is easy. Java is powerful.
Characters: 31
Words: 6
Sentences: 2
Editing History:
```

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```
18         if(!s.trim().isEmpty()){
19             sentenceCount++;
20         }
21     }
22
23     String oldWord=sc.next();
24     String newWord=sc.next();
25
26     String edited=paragraph.replace(oldWord,newWord);
27
28     StringBuffer history=new StringBuffer();
29     history.append("Replaced ").append(oldWord).append(" With ").append(newWord);
30
31     StringBuffer finalDoc=new StringBuffer();
32     String arr[]=edited.split("[.!?]");
33
34     for(String se:arr){
35         String text=se.trim();
36         if(!text.isEmpty()){
37             finalDoc.append(new StringBuffer(text).reverse().append(" "));
38         }
39     }
40
41     System.out.println("Original Document:");
42     System.out.println(paragraph);
43
44     System.out.println("Characters: "+characters);
45     System.out.println("Words: "+wordCount);
46     System.out.println("Sentences: "+sentenceCount);
47
48     System.out.println("Editing History:");
49     System.out.println(history);
50
51     System.out.println("Final Document:");
52     System.out.println(finalDoc);
53 }
```



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```
31     StringBuilder finalDoc=new StringBuilder();
32     String arr[]=edited.split("[.!?]");
33
34     for(String se:arr){
35         String text=se.trim();
36         if(!text.isEmpty()){
37             finalDoc.append(new StringBuilder(text).reverse()).append(" ");
38         }
39     }
40
41     System.out.println("Original Document:");
42     System.out.println(paragraph);
43
44     System.out.println("Characters: "+characters);
45     System.out.println("Words: "+wordCount);
46     System.out.println("Sentences: "+sentenceCount);
47
48     System.out.println("Editing History:");
49     System.out.println(history);
50
51     System.out.println("Final Document:");
52     System.out.println(finalDoc);
53
54     sc.close();
55 }
56 }
```

 

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10 marks

Q5
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```
1 import java.util.*;
2
3 class Patient {int id;String name;String department;
4
5 Patient(int id, String name, String department) {
6     this.id = id;
7     this.name = name;
8     this.department = department;
9 }
10
11 public static void main(String[] args) {
12
13     Scanner sc = new Scanner(System.in);
14     List<Patient> patients = new ArrayList<>();
15
16
17     System.out.print("Enter number of patients: ");
18     int n = sc.nextInt();
19     sc.nextLine();
20
21     for (int i = 0; i < n; i++) {
22         System.out.println("\nEnter Patient " + (i + 1) + " Details");
23
24         System.out.print("Patient ID: ");
25         int id = sc.nextInt();
26         sc.nextLine();
27
28         System.out.print("Name: ");
29         String name = sc.nextLine();
30
31         System.out.print("Department: ");
32         String dept = sc.nextLine();
33
34         patients.add(new Patient(id, name, dept));
35     }
36 }
```

Orthopaedics
103
Cardiology

Output
Enter number of patients:
Enter Patient 1 Details
Patient ID: Name: Department:
Enter Patient 2 Details
Patient ID: Name: Department:
Enter Patient 3 Details

```
35 }
36
37
38 System.out.println("\nPatients in Registration Order:");
39 for (Patient p : patients) {
40     System.out.println(p.id + " " + p.name + " " + p.department);
41 }
42
43
44 System.out.print("\nEnter Patient ID to Update: ");
45 int updateId = sc.nextInt();
46 sc.nextLine();
47
48 for (Patient p : patients) {
49     if (p.id == updateId) {
50         System.out.print("Enter New Name: ");
51         p.name = sc.nextLine();
52
53         System.out.print("Enter New Department: ");
54         p.department = sc.nextLine();
55         break;
56     }
57 }
58
59
60 System.out.print("\nEnter Patient ID to Remove: ");
61 int removeId = sc.nextInt();
62
63 patients.removeIf(p -> p.id == removeId);
64
65 sc.nextLine();
66
67
68 System.out.print("\nEnter Department to Display: ");
69 String dept = sc.nextLine();
```

```
66
67
68 System.out.print("\nEnter Department to Display: ");
69 String dept = sc.nextLine();
70
71 System.out.println("\nPatients in " + dept + " Department:");
72 for (Patient p : patients) {
73     if (p.department.equalsIgnoreCase(dept)) {
74         System.out.println(p.id + " " + p.name);
75     }
76 }
77
78
79 if (!patients.isEmpty()) {
80     System.out.println("\nFirst Registered Patient:");
81     Patient first = patients.get(0);
82     System.out.println(first.id + " " + first.name);
83
84     System.out.println("\nLast Registered Patient:");
85     Patient last = patients.get(patients.size() - 1);
86     System.out.println(last.id + " " + last.name);
87 }
88
89 sc.close();
90 }
91
92 }
```

Run

Save

using Set Interface
10 marks

Q5

Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```
1 import java.util.*;
2
3 public class AirportBoardingQueue { public static void main (String[] args) {
4
5     Queue<String> queue = new LinkedList<>();
6     PriorityQueue<String> vip = new PriorityQueue<>();
7
8     int n = sc.nextInt();
9     sc.nextLine();
10
11     for (int i = 0; i < n; i++) {
12         queue.add(sc.nextLine());
13     }
14
15     System.out.println("Current Queue:");
16     for (String s : queue) {
17         System.out.println(s);
18     }
19
20     if (!queue.isEmpty()) {
21         System.out.println("Next Passenger:");
22         System.out.println(queue.peek());
23     }
24
25     int v = sc.nextInt();
26     sc.nextLine();
27
28     for (int i = 0; i < v; i++) {
29         vip.add(sc.nextLine());
30     }
31
32     System.out.println("VIP Queue:");
33     while (!vip.isEmpty()) {
34         System.out.println(vip.poll());
35     }
36 }
```

Ravi
2
Vip1
Vip2

Output

```
Current Queue:
Rahul
Anita
Kiran
Ravi
Next Passenger:
```

```
24
25 int v = sc.nextInt();
26 sc.nextLine();
27
28 for (int i = 0; i < v; i++) {
29     vip.add(sc.nextLine());
30 }
31
32 System.out.println("VIP Queue:");
33 while (!vip.isEmpty()) {
34     System.out.println(vip.poll());
35 }
36
37 System.out.println("Boarding Passengers:");
38 while (!queue.isEmpty()) {
39     System.out.println("Boarded: " + queue.poll());
40     System.out.println("Remaining Queue:");
41     for (String s : queue) {
42         System.out.println(s);
43     }
44 }
45
46 sc.close();
47 }
48
49 }
```

Run

Save

Q5
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```
1 java.util.*;
2
3 class DigitalCertificateVerification{
4     static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6
7         LinkedHashSet<String> certificates=new LinkedHashSet<>();
8
9         int n=sc.nextInt();
10        sc.nextLine();
11
12        for(int i=0;i<n;i++){
13            certificates.add(sc.nextLine());
14        }
15
16        System.out.println("Certificates in Insertion Order:");
17        for(String s:certificates){
18            System.out.println(s);
19        }
20
21        System.out.print("Enter Certificate ID to Verify: ");
22        String verify=sc.nextLine();
23
24        if(certificates.contains(verify)){
25            System.out.println("Certificate Exists");
26        }else{
27            System.out.println("Certificate Not Found");
28        }
29
30        TreeSet<String> sorted=new TreeSet<>(certificates);
31
32        System.out.println("Certificates in Sorted Order:");
33        for(String s:sorted){
34            System.out.println(s);
35        }
36    }
37 }
```

5
C101
C102
C103

Output

```
Certificates in Insertion Order:
C101
C102
C103
C104
Enter Certificate ID to Verify:
```

```
26 }else{
27     System.out.println("Certificate Not Found");
28 }
29
30 TreeSet<String> sorted=new TreeSet<>(certificates);
31
32 System.out.println("Certificates in Sorted Order:");
33 for(String s:sorted){
34     System.out.println(s);
35 }
36
37 System.out.println("Total Unique Certificates: "+certificates.size());
38
39 System.out.print("Enter Certificate ID to Remove: ");
40 String remove=sc.nextLine();
41
42 certificates.remove(remove);
43
44 System.out.println("Certificates After Removal:");
45 for(String s:certificates){
46     System.out.println(s);
47 }
48
49 sc.close();
50
51 }
```

Run

Save

Q5
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```
1 import java.util.*;
2
3 class Product{int id;String name;String category;int quantity;double price;}
4
5 Product (int id,String name,String category,int quantity,double price){
6     this.id=id;
7     this.name=name;
8     this.category=category;
9     this.quantity=quantity;
10    this.price=price;
11 }
12
13 public static void main(String[] args){Scanner sc=new Scanner(System.in);
14
15     int n=sc.nextInt();
16     sc.nextLine();
17
18     for(int i=0;i<n;i++){
19         int id=sc.nextInt();
20         sc.nextLine();
21         String name=sc.nextLine();
22         String category=sc.nextLine();
23         int quantity=sc.nextInt();
24         double price=sc.nextDouble();
25         sc.nextLine();
26
27         map.put(id,new Product(id,name,category,quantity,price));
28     }
29
30     int updateId=sc.nextInt();
31     int newQuantity=sc.nextInt();
32
33     if(map.containsKey(updateId)){
34         map.get(updateId).quantity=newQuantity;
35     }
```

3
101
Laptop
Electronics

Output

```
102 Laptop
102 Mouse
Available Products:
101 Laptop Electronics 5 50000.0
102 Mouse Electronics 15 800.0
```

```
36
37 int searchId=sc.nextInt();
38
39 if(map.containsKey(searchId)){
40     Product p=map.get(searchId);
41     System.out.println(p.id+" "+p.name+" "+p.category+" "+p.quantity);
42 }else{
43     System.out.println("Product Not Found");
44 }
45
46 double total=0;
47
48 for(Product p:map.values()){
49     total+=p.quantity*p.price;
50 }
51
52 System.out.println("Total Inventory Value: "+total);
53
54 sc.nextLine();
55 String category=sc.nextLine();
56
57 System.out.println("Products in "+category+":");
58
59 for(Product p:map.values()){
60     if(p.category.equalsIgnoreCase(category)){
61         System.out.println(p.id+" "+p.name);
62     }
63 }
64
65 int removeId=sc.nextInt();
66 map.remove(removeId);
67
68 System.out.println("Available Products:");
69
70 for(Product p:map.values()){
71     System.out.println(p.id+" "+p.name+" "+p.category+" "+p.quantity);
72 }
```

```
52 System.out.println("Total Inventory Value: "+total);
53
54 sc.nextLine();
55 String category=sc.nextLine();
56
57 System.out.println("Products in "+category+":");
58
59 for(Product p:map.values()){
60     if(p.category.equalsIgnoreCase(category)){
61         System.out.println(p.id+" "+p.name);
62     }
63 }
64
65 int removeId=sc.nextInt();
66 map.remove(removeId);
67
68 System.out.println("Available Products:");
69
70 for(Product p:map.values()){
71     System.out.println(p.id+" "+p.name+" "+p.category+" "+p.quantity);
72 }
73
74 sc.close();
75 }
76
77 }
```

 Run Save